



## Ancient Vases in 3D: Digitisation, Research and Web-based Publication

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**Keywords:** 3D Digitisation, Photogrammetry, Ancient Vases, Digital Cultural Heritage, Online Publication, Museum Collection Online, Badisches Landesmuseum Karlsruhe (BLM)

Among the most distinguished holdings of our museum is a collection of approximately 550 ancient Greek and South Italian vases, with particular strengths in Late Archaic and Classical Attic black- and red-figure pottery and Apulian red-figure ceramics. Partially acquired already during the early nineteenth century, the collection forms the core of the museum's antiquities holdings and represents an important resource for archaeological research.

The project, funded by the German Research Foundation (DFG), combines the scholarly reassessment of approximately 300 selected objects with high-resolution 2D photography, colour-calibrated photogrammetric 3D digitisation, and the development of a sustainable web-based publication infrastructure. Rather than treating digitisation as an end in itself, the project investigates how high-quality 3D data can be integrated into a research-oriented digital collection that supports documentation, scholarly analysis, and public engagement.

The project addresses three central research questions: (1) How can established archaeological research methods be enhanced through high-resolution 2D and 3D digital surrogates? (2) What additional research potential does three-dimensional documentation provide for the study of production techniques, traces of use, and object biographies? (3) How can digital collections be designed to be simultaneously accessible to both specialist and non-specialist audiences?

Methodologically, the project is based on a colour-calibrated photogrammetric workflow that has been adapted to the specific characteristics of archaeological ceramics. Highly reflective black-gloss surfaces, homogeneous colour areas with limited image features, and the documentation of vessel interiors through narrow necks require customised acquisition strategies, including optimised lighting configurations and carefully controlled camera geometry for the object-specific image acquisitions. These adaptations are

evaluated with regard to geometric accuracy, texture quality, and reproducibility within a scalable digitisation workflow.

A central outcome of the project is a responsive online collection published on the website of the Badisches Landesmuseum Karlsruhe (BLM) in which each object combines scholarly catalogue information, high-resolution images, and interactive 3D models. Metadata are structured according to established cultural heritage standards and linked to persistent identifiers to facilitate interoperability with national and international research infrastructures, including NFDI4Culture and NFDI4Objects. The publication workflow alligns with FAIR and CARE principles while supporting Linked Open Data approaches that enable long-term reuse, citation, and integration with external archaeological resources.

The web platform incorporates an annotation-enabled 3D viewer offering measurement tools, adjustable lighting, sectional views, and semantic annotations directly linked to object metadata. This enables researchers to document observations transparently while providing accessible visualisations for museum visitors, teaching, and digital exhibitions. By integrating documentation, metadata, and interactive visualisation within a single publication environment, the project contributes to current discussions on digital heritage infrastructures.

Beyond the creation of high-quality digital surrogates, the project aims to establish a reproducible workflow for the scholarly publication of archaeological 3D data, demonstrate how digital documentation can generate new research perspectives on ancient ceramics, and provide a reusable framework for comparable museum collections.



*Fig. 1 The 3D modelling procedure of the 3D4Vases Project held by the Badisches Landesmuseum Karlsruhe. Photo: Hannes Deters.*

## References

1. Hafner, G. (1951), *Corpus Vasorum Antiquorum Karlsruhe 1, Deutschland 7*, Verlag C. H. Beck.
2. Hafner, G. (1952), *Corpus Vasorum Antiquorum Karlsruhe 2, Deutschland 8*, Verlag C. H. Beck.
3. Weiss, C. (1990), *Corpus Vasorum Antiquorum Karlsruhe 3, Deutschland 60*. Verlag C. H. Beck.
4. Maaß, M. (2007), *Maler und Dichter: Mythos, Fest und Alltag. Griechische Vasenbilder aus der Sammlung des Badischen Landesmuseums Karlsruhe*.